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A design-based approach to analysing student engagement with a GenAI-Enabled brainstorming app☆

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ABSTRACT

While there are several "writing buddy" Generative Artificial Intelligence (GenAI) apps that check grammar and language usage, not many focus exclusively on enhancing the brainstorming process for writing across disciplines. To fill this gap, a team of staff and student assistants with programming and User Interface (UI) and User Experience (UX) expertise designed and prototyped a web app named "Waai," which rhymes with 'why,' that could assist students throughout the writing process for a first-year general writing module for all undergraduate students at a Singaporean university.

Utilising surveys, focus group discussions, and app data that shows the nature and type of student engagement with the Waai app, this paper studies the impact of one aspect of the Waai app, an uni-directional chatbot named "Nudgy," as a first step to optimising interactions with an AI chatbot for writing purposes. Overall, we found that students were able to benefit from the GenAI chatbot Nudgy in 5 distinct ways: 1) its pre-engineered prompts, which were tailored to the course assignment rubrics; 2) its tendency to recommend topics to research rather than give students answers; 3) its suggested research topics, which helped students to consider different perspectives on their topics; 4) how it modelled ways to ideate new insights; and 5) its constant availability. Students, however, expressed reservations about the Nudgy, particularly in terms of: 1) the limitations of pre-engineered prompts within the app; 2) difficulty in discerning the most relevant of the Nudgy feedback; 3) mistrust in GenAI and Aigiarism; and 4) a recognition of the limitations of GenAI in supporting argumentative writing.

"Waai" essentially presents a decision-making framework for brainstorming based on cognitive socialisation, a method of learning that emphasises inductive as opposed to deductive experience that could be applied to online environments, as an ideology of learning that considers, among other aspects, the development of selfhood, where learning is both guided and mediated (Kesebir & Gardner, 2010). In the context of asynchronous learning, meaning is not intrinsic but rather picked up through interactions on online platforms. Interacting with a chatbot with pre-designed prompts result in a ritual that both define and explore the limits of knowledge building. Symbolic interactionism (Aksan et al., 2009) through the medium of technology is a key objective of 21st century education, where 'learning' is internalised as an individual experience. Waai offers educatros additional understanding of the effects of personalization (Mygland et al., 2021) as proposed by this design-based study of the role of instruction in the creation of online 'learning' experiences. The centrality of instruction and standards of reasoning through the process of brainstorming suggests that the developmental stages of 'learning' concepts could empower a process for self-regulation (Zimmerman, 1989) that goes beyond immediate causes and effects to inspire a spiral of reflection and change essential to ideation.

1. Introduction

A team from a Singaporean university comprising of faculty and student assistants from different faculties envisioned a chatbot that would "nudge" students in new directions or paths by providing personalized feedback to support active brainstorming and even self-evaluation through the use of GenAI, in particular, ChatGPT 3.5 and 4. This app, called Waai, consists of an uni-directional chatbot and was

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completed and piloted in 2023–2024, with approximately 130 students over two semesters, for an inquiry-based writing module for all undergraduate students at the university.

By challenging the myth of intrinsic technological effects (Lodge, Thompson, & Corrin, 2023) as well as promoting the democratisation of knowledge in an unprecedented 'knowledge' age, Waai was used as part of a pedagogical sequence of in-class activities instead of a tool in isolation. To complement the training of language skills during tutorials, Waai includes a word association chart to encourage the use of figurative language to think outside the box. This was based on the belief that contrasting ideas could work to generate original writing, where writing is seen to be a process of thinking.

Waai employs design-based digital storytelling that has been studied for its use in the instructional design of language skills teaching, as well as product promotion (Ramalia, 2023). The creation of a digital narrative suggests how technology is not just a means to an end but rather a point of deeper reflection in relation to enabling "transformational" or conceptual learning from the learner's point of view (Hessler & Lambert, 2017; Meyer et al., 2010). The objective of writing might be seen as a journey to gain entry into a particular discourse of knowledge, rather than to arrive at expertise. In this way, "transformational learning" is associated with learning something new. The importance of the process of writing as a lived experience considers the iterative cycles of thinking and conceptual reckoning (Meyer & Land, 2005) involved in the transformation of learning even outside of the classroom. While this approach to writing is a liberation of learning through the discovery of authenticity (Freire & Ramos, 1970), it also requires necessary accountability in relation to the teacher's role in framing a learner-centric writing pedagogy.

Burnett and Merchant (2020) in Undoing the Digital (2020) have argued for a shift from viewing literacy as merely foundational, that is, as a "literacy event", to a consideration of "literacy as event". The design thinking framework is used to introduce an iterative process of brainstorming beginning with empathising with their observations and refraining from making a judgement. Students are also encouraged to brainstorm ideas, both good and bad. In this way, the app helps shape student learning and indirectly assesses their understanding of the different components of the writing process, which begins with fieldwork to ensure divergent thinking. This inquiry framework requires students to identify a genuine or authentic problem that avoids making a stand. This problem is derived from selecting and analysing observations from fieldwork.

Waai prompts users to put on different disciplinary lenses that inform their interpretation of sources. Waai does not automatically search out sources for students but offers a two-level framework that consists of the subject area of research (the 'field of study') and the substantial concepts to be deconstructed. We also focused on how Waai users might approach their individual fieldwork experiences by distinguishing implicit and explicit observations and eventually how they might bring various concepts into play in their developing argument. In order to realize each student's individualized argument through GenAI, the analysis of different information and concepts is accumulated in a sequence or flow using an iterative process inspired by design thinking.

This paper argues that the skill of brainstorming involves a willingness to face difficulty (Lodge et al., 2018), a mindset essential to self-regulation (Zimmerman, 1989) that Waai encourages through its backend prompting framework that is driven by people behind the scenes rather than only an AI bot. In this regard, this paper adopts A Design Thinking Machine Learning framework (Cochrane, 2017) that focuses on developing student self-efficacy through the process of scaffolding the writing process in a built-from-scratch full-fledged app that uses GenAI. To study the potential for 'writing presence' as an alternative to "teacher presence" (Stanier, 2015) through a controlled process of GenAI engagement using the Nudgy chatbot, we explore affectivity as a mindset to learning that is essential to self-regulation (Zimmerman, 1989) and may be promoted by adopting the non-linear process of design thinking grounded on prototyping.

Although framed as a sequential process, Waai uses design thinking for a more flexible exploratory structure students to approach a new problem. The intellectualisation of ideas is emphasized, through the integration of the anticipatory backend prompting and user journey experience that is significantly marked by uncertainty. As the following excerpt from the initial ideation of the user journey experience suggests through the medium of emoticons, the purpose of Waai is to help students bounce back emotionally, thereby increasing attitudinal task facility by at least bringing things back to equilibrium. The use of educational technology is seen to be potentially useful in accompanying students on their individual journeys as they each explore a new topic from scratch.

Yet, the use of LLMs is fraught with inevitable limitations due to how it is trained on existing data and textual patterns, leading some to suggest that it may not be possible to expect that GenAI chatbots is capable of actually connecting ideas. With this knowledge in the background, Waai thrives precisely because it offers a middle-ground between "generative" and "predictive" AI (Narayanan & Kapoor, 2024) where machine learning is only implemented where thought to be useful, despite the limitations of GenAI as a creative tool. In short, it is both pedagogically and technologically sound that the design of Waai privileges divergent thinking over self-determinism. In the desire to use technology to promote individual student voices, we believed the combination of ideas from ChatGPT-4 as an advanced search engine and the randomisation of human associations could result in greater conceptual engagement essential in the process of writing *as thinking*.

Thus, our research question was: When brainstorming in the writing process, how do students perceive and interact with GenAI that is mediated through pre-engineered prompts?

1.1. Literature review

Artificial Intelligence (AI), together with its subset, machine learning, is not a recent development but one that originated in 1943. In the 1980s, it became a way of analyzing and interpreting large amounts of data based on existing computer programs like Microsoft Excel. For a long time, machine learning has been recognized as an aid in problem-solving, its ability to do so dependent on the amount and quality of data that its algorithm has access to. Thus, until recently, machine learning has been thought to be limited in that it still relies on human-generated data (Pichler & Hartig, 2023).

AI was generally considered to be limited to data analysis until the famous release in November 2022 of a generative AI algorithm called GPT-3, first made available in 2020. In November last year, it became accessible without cost to the user. The focus on ever-larger large language models (LLMs) in English has become a driver to understand the risks involved, but also the seeming benefits of predictive technology in generating new knowledge (Brooks, 2023).

GenAI is increasingly regarded as a highly rated search engine whose accuracy is nonetheless seen as having reliability issues precisely because LLM-generated answers seem "persuasive" (Stokel-Walker & Van Noorden, 2023). Apart from reliability, some have highlighted the lack of specificity despite the ability of AI to summarize an article and even brainstorm ideas and therefore transform our knowledge bases. As with all research, it becomes important to test it for biases and even deliberate falsity in terms of the knowledge it generates (Bender et al., 2021; Stokel-Walker & Van Noorden, 2023). These issues of specificity, bias and falsity in relation to AI-generated information can pose a problem for learners in academic and technical research settings. Nonetheless, it is the known limitations of GenAI that might help us focus on what's important in activating open-ended inquiry beyond transactional uses of AI (Lodge, Yang, Furze, and Dawson, 2023). This especially applies to tasks where new context is needed and there is no AI trained to predict the next step (Brooks, 2023; Dagostino, 2023).

While it is generally acknowledged that ChatGPT is a tool of the future, ChatGPT is currently not thought to be consistently accurate in relation to scientific or technical topics. Even though there are testimonials that ChatGPT could help researchers in scientific disciplines distill evidence for lab work, thus making the research process more efficient, a human moderator is still necessary to ensure possible inaccuracies are weeded out (Salvagno et al., 2023). Yet, larger LLMs like Google's Minerva AI can employ inference based on "chain of thought" such that it might even appear that "reasoning" can also be trained. The instructive word is that the AI *appears* to be able to reason but all evidence appears to point to how it is not as "systematic" or rigorous as human thought and its lapses in judgement and understanding are decidedly non-human. For instance, it has been argued that machine-learning-based systems cannot differentiate causation and correlation. Where learning is prescribed rather than evolving, the result is that "underlying causes" are left unexamined (Gillani, N. et al., 2023).

At the same time, the key concern in academia where plagiarism is treated as a matter of academic misconduct, is in determining whether AI-detection software works. Turnitin, for instance, has been working on a solution for users on behalf of schools, universities and scholarly publishers worldwide who use its products, while, GPTZero is an AI-detection tool that identifies particular features of an AI-generated text such as "perplexity" or the predictability of a text, and "burstiness" or a cadence that tends to be less varying than text written by humans (Stokel-Walker & Van Noorden, 2023). Yet, none of these tools would be effective if there is any human intervention in the AI-generated texts. Thus, watermarking AI outputs has been seen as one possible AI-proofing strategy (Hern, 2022; Stokel-Walker & Van Noorden, 2023).

Thus, it has been argued that educators cannot leave understanding of language models to those in computer science. Decisions about the amount of reliance on AI assistance allowed in assignments could result in changes in terms of the way students are assessed. Students also need to be briefed about expectations and ground rules where AI assistance could help achieve an intended outcome more effectively. This could impact the consistency in which academic misconduct is treated across the disciplines (Dagostino, 2023).

While "Aigiarism" or AI-assisted plagiarism (Mustafa, 2024) is seen as an immediate threat in academic circles, AI chatbots are becoming more advanced in terms of their ability to predict not just text but extended conversations in an almost authentic manner (Hern, 2022). The fact that ChatGPT is a wealth of information suggests that it cannot be ignored. The solution, for most educators, appears to be in ensuring that assignments are tweaked such that they are more deeply personal and less generic (Wallbank, 2023). Whether this could transform what teaching writing might consist of remains to be seen (Potter, 2020).

1.1.1. GenAI and writing assignments

There is a danger that students can reproduce a decent piece of writing without any authentic engagement with ideas. Writing is a tool for thinking; if we offload writing to GenAI, we will be sabotaging the thinking process. Thus, educators realize that a revaluation of teaching assessment proficiency is necessary. While Wallbank (2023) has suggested a focus on dialogue rather than the final product, Brooks (2023) has pointed out that GenAI "shows us what it can't do" which might indirectly help us know where to channel our creative energies in writing contexts.

Assignment design should focus on process, revision, nuanced thinking and lived experience. Topics should be based on choices made by students. In addition, as instructors, we need to appeal to students by emphasizing *why* they need to learn to think and write. The dilemma of AI as both a threat to traditional standards of assessment and a tool that could be used to improve instruction by doing less is apparent in academia. Gillani et al. (2023), for instance, has argued that the "meta learning" of machine learning "generally lacks the higher-order thinking, reflection, and planning woven throughout human meta-learning" (pg. 106). However, we argue that this also suggests possibilities in terms of customization when GenAI is used within a more controlled environment of an app designed specifically to teach a certain skill.

An assignment that can deal with the advent of ChatGPT is one that focuses on the process that students go through rather than only on the product they produce, though the latter would still have to be graded. This has led to faculty building their own FAQ chatbots built on an RAG system. At our university, this has included Project NALA (Lai et al., 2024), among others. They target specific audiences within their schools and largely are designed to meet specific user needs by targeting accuracy of information based on the premise that GenAI can help streamline the process of information dissemination, especially for large classes.

Further afield, writing apps tend to offer editing, with only a select few offering prompts that actually support student writing during the difficult process of revision. In particular, the groundbreaking FLoRA Project (FLoRA Project), a research study around a writing app platform originating in Monash University and the product of a collaboration between Bannert, M., Gašević, D., & Molenaar, I. (Molenaar et al., 2023), uses scaffolded prompts based on Bloom's Taxonomy for users to engage with different parts of the writing process. As there are several layers to the platform, offering both free and fixed prompting, it grants students the freedom to input prompts that might be general to the piece of writing on the whole but at the same time aims to develop student awareness at each step of the writing process (Lim et al., 2021; Van Der Graaf et al., 2021; Fan et al., 2022). Allowing for free prompting alone might not be a significant add-on to what students are already able to do on free GenAI platforms like ChatGPT. For Waai (Waai), pre-designed prompts, together with a backend prompting system, were used to promote unidirectional conversations with the chatbot to clarify contextual awareness at each step of the user journey as well as a sense of ownership over individual choices made in a piece of writing (Kılıçkaya, 2020), without granting the possibility of free prompting.

As Hwang and Chen (2023) claim, "learners benefit from ChatGPT based on the way they interact with it". In particular, they argue that "properly adopting GenAI applications in learning designs could shift technology-enhanced learning to a different level, in which students can fully exploit their creativity and application capabilities to create artworks, solve problems, or complete projects with the assistance from GAI" (Hwang & Chen, 2023). Hwang and Chen (2023) warn that if educators simply ban GenAI, students might be more inclined to use such tools in ways that could compromise the integrity of their work such as submitting AI-generated writing as their own. In short, they argue that learning design is important where GenAI is concerned. Otherwise, the future may truly lie with students who are more advanced users of prompt engineering. Apps that offer built-in contextualisation, like Waai, hope instead to level the playing field.

1.2. The current study

The Waai web app feeds student writing input into ChatGPT-4 and moderates output using pre-designed prompts and prompt templates including a system or meta-prompt that asks the app to be a 'writing assistant'.

We aimed to create a chatbot that demonstrates a certain level of experience without establishing a hierarchy of knowledge that Paulo Freire (1970) once described in relation to the concept of "banking" where teacher-led instructional outcomes could lead to education as "an instrument of oppression" (pg. 72). In an age of AI, it has become even more essential to ensure that education does not revert to "an act of depositing" (Freire & Ramos, 1970; pg. 72) which is aggravated by authoritative-sounding chatbots that might perpetuate an underlying teacher-centred pedagogy despite the purported aim of GenAI to involve more student-centred and even independent learning. In other words, we are alert to the possibility of negating the very process of inquiry that we hope to encourage and initiate within the app.

Thus, Waai uses a recursive framework that offers a supportive network of ideation for students where the use of GenAI is already widespread especially with the introduction of publicly available free tools like ChatGPT-3.5 and ChatGPT-4o. In this regard, a visual table (see Table 1) based on the Technology Acceptance Model (Davis, 1989) to study acceptance and actual usage of technological tool responds to reservations in relation to GenAI use and expectations.

Importantly, the hallucinations of AI, because of the nature of its brainstorming objective, would not be a hindrance in that the responsibility for looking up concepts and assessing their credibility, relevance, and significance still lies with students. This is where Waai is complemented by peer review exercises in the classroom on assignment drafts that employ critical thinking elements of thought, as well as one-on-one consultations with the writing instructor based on a single-point rubric containing question prompts.

Waai is designed to encourage brainstorming. By dissecting an entire essay into different steps, the chatbot design ensures that students' responses are mediated by an inbuilt problem-solving process where student writing at each stage is considered separately and then together, through the accumulation of chat histories from chosen prompts, which feed student writing to ChatGPT-4 through the use of a system prompt. The chatbot design results in suggestions rather than prescribed answers. Data is saved at each stage, including the theme selected for their essay, their personal interests, as well as current inspirations, to ensure their lived experience is captured. This personalization in Waai is one of the affordances in human-chatbot interaction (Mygland et al., 2021), which offered an opportunity to deliberately engineer GenAI output to avoid overdependence on AI in order to preserve the individual writing voice.

The ongoing co-creation of the app, between faculty and students, offers a glimpse of the potential for research into social cognition in the evaluation of technology in higher education. This reveals insights into social constructivism as a model for the development of critical thinking as a key result of a 21st century education whose necessity has only accelerated in the face of the ubiquity of GenAI, especially in relation to editing in writing (Hosseini et al., 2023).

The interface of the chatbot, instead of simulating a "live" conversation, captures students' ongoing process of brainstorming and ideation, across the different pages where the chatbot is available. Using predesigned prompts based on curriculum, the potential of AI is maximised for users who are guided to revise, clarify, and improve their ideas through a series of different interactions. The environment of this app occurs within certain constraints which we hoped would become the catalyst for knowledge building through modelling choice. The provision of choice would be made possible by the built-in front-end prompts via the chatbot function on most pages of the app (see Fig. 2), as well as the backend prompt engineering including the use of a system prompt. However, users of the app would not be interfacing directly with what happens on the backend. Instead, the prompts shape their immersion into the writing space that Waai supports, as the following flow chart (see Fig. 3) demonstrates.

Generating qualitative and quantitative data from the app, surveys, and focus group discussions, offered the empirical evidence necessary to understand the impact of GenAI on the intended learning outcomes of the writing module (Weng et al., 2024). This paper could be useful to educators in relation to both the development and implementation of an educational app prototype that is from scratch which could be translated to other disciplines and even possibly scaled for other research projects that focus on concepts. Importantly, this paper acknowledges the necessity of students being involved in the creation process from the start, as well as the continual involvement of students in the building up of this space.

Table 1
Summary of considerations in the development of a GenAI app in higher education.

Steps to successful implementation of GenAI tool	Discovery Why do students use GenAI?	GenAI usage Why would students trust the tool?	Student Self-Sufficiency How can students feel successful as a user of GenAI?
Actions What does the student do?	Use LLMs outside of class or even in class to improve task efficiency.	The app has been designed by instructors of the course.	Watch an onboarding video.
		The app is hosted on an enterprise platform.	Explore the chatbot on their own. Explore examples of how each page could be filled in.
Needs and Pains What does the student want to achieve or avoid?	The student wants to achieve better quality ideas.	The student appreciates how the pre-designed prompts mirror the language of the course curriculum and its intended goals.	The student does not have to perform prompt engineering on their own in interacting with a GenAI chatbot.
	The student wants to avoid Aigiarism due to rules in relation to GenAI use for school assignments.	The student is assured of the privacy of their data.	A video can help students get oriented easily on a new platform.
		The student knows the app is not just free but rather levels the playing field for everyone.	The student may be inspired by how others have used the app if there is a community on the platform to exchange ideas.
Opportunities What should app designers build or improve?	Increase the likelihood of students signing up by creating a student dashboard to track usage. Do they want ease of access or high value? How might we make a compromise?	Increase resources for brainstorming by creating a collaborative platform that invites a community of experts and encourages peer feedback.	Onboard new users quickly by implementing a video introduction.

1.2.1. Pedagogical sequencing for the introduction of mobile technology

The design of Waai takes student users through their entire process of brainstorming (see Fig. 4). For this first-year writing course, this process entails that students first observe a place or community, writing down their observations as Fieldnotes, which they submit for peer and teacher feedback. Second, students submit a marked first assignment that contains a focused and vivid description of selected observations, an analysis of patterns and particular details they found intriguing, culminating in a conceptual question that springs from their observations. After receiving teacher feedback on this assignment, students revise and move forward to finding two sources that can help them respond to their research question. In their third submission of the semester, students share summaries and evaluations of their chosen sources as Research Notes. After receiving teacher feedback, students assemble their first draft, which receives peer and teacher feedback as they revise toward their final written argumentative essay.

Waai's Nudgy differs from other chatbots as it should not be used as a stand-alone chatbot but rather as part of a larger sequence of digital and non-digital activities planned around specific learning outcomes (see Fig. 5). We integrated GenAI into the classroom yet made it transportable as well for early steps in the scaffolded writing process. As opposed to a writing desktop app where one might feed student writing into the chatbot to get its responses, we focused on extending the brainstorming

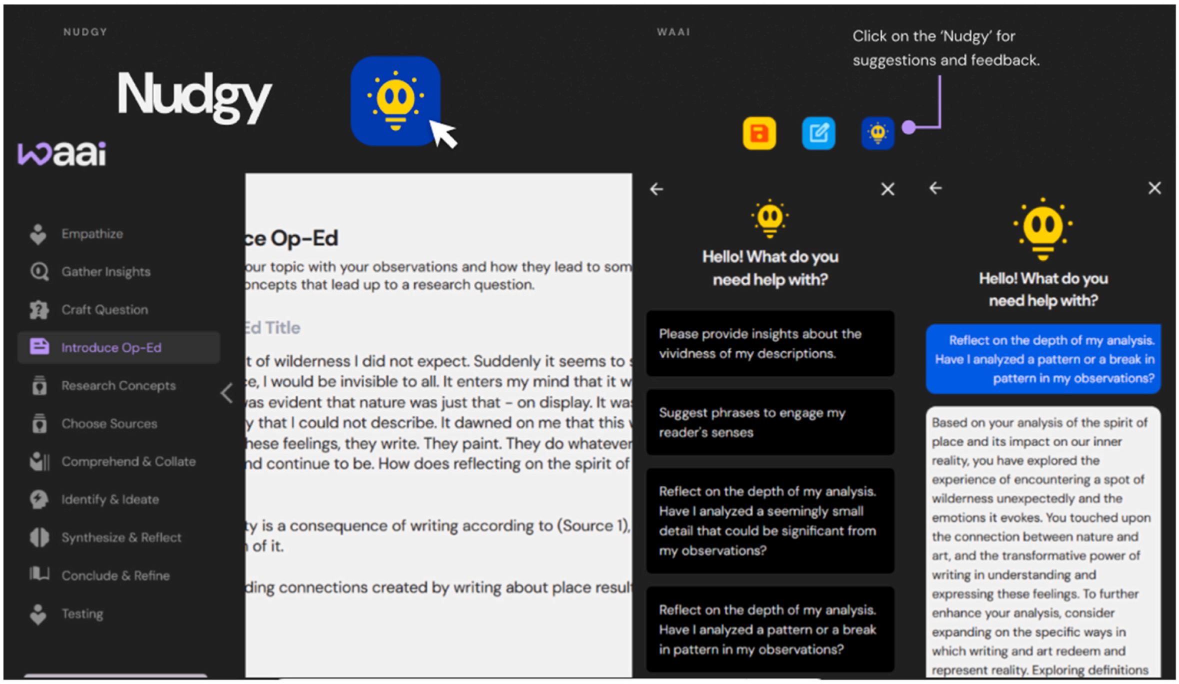

Fig. 2. Sample Waai Chat Interface on "Introduce Op-Ed" page.

Fig. 3. Simplified meta-prompt flow in Waai.

capacity of GenAI by making AI an inclusive rather than exclusive tool for thinking. Cope et al. (2020) highlights that the naming and clarifying of concepts is essential to AI-enabled learning. Due to its vast "Lookup-ability, the use of AI in education mandates capacity building in relation to the possibility of developing "a collaborative mnemonics" that go beyond adopting a single voice or opinion (6). Thus, GenAI feedback output was just one set of feedback students received, as seen in the following diagram.

Waai employs a three-step approach of 1) providing raw material in the form of fieldwork observations, 2) processing input through an interdisciplinary lens, and 3) identifying connections based on student input to generate ideas, concepts, and phrases, offering a cyclical approach where students are able to self-evaluate and consider the proposed concepts, ideas, and phrases as they revise. The prompt engineering in Waai doesn't currently bridge the gap between the knowledge of these concepts and ideas and their actual application. Choices in terms of ideas and actual conceptual integration are left entirely up to the student writer. Thus, the validation mechanism relied on human testing of the prompt templates and focused less on assessing accuracy of judgement but rather on encouraging actual use.

1.2.2. *Design framework for mobile learning*

Based on a Design for Transformative Mobile Learning *Framework (Cochrane* et al.*, 2017, p. 27),* "Waai" might be framed in relation to how technology functions along cognitive, metacognitive, and epistemic levels to support personalized learning.

The conception of ideas for a student Op-Ed is supported at two levels in Waai – through the prompt design as well as the act of writing as social cognition where the chatbot theoretically helps students go beyond factual or descriptive writing. In the context of "Waai", the prompts were designed to ensure that the primary function of the app is for brainstorming. In designing the app, we are not simply looking for a solution or even an answer to their research question. The structure of the entire app is designed to empower human learning. We adopt a 'human-in-the-loop' system through a sequence of interconnected pages.

The promotion of the writer's agency by the Nudgy chatbot is implied at best through pre-designed prompts that require a level of reflection and immediacy that the chatbot presentation in ChatGPT has been said to offer, yet retaining control over what help is being provided (Mactear et al. (2016). Right now, these prompts are parked across different pages to precipitate design thinking through the ideation process for students. These building blocks of writing lead to more abstract, iterative processes (with a capacity to form or entertain ideas) associated with research-based writing. In addition, using the pronoun 'you' in formulating the prompts anticipates a transformative act of questioning.

In this vein, the DTML framework *(*Cochrane et al., 2017*)* allows us to consider how Waai might inspire "creativity" at different levels – beyond "reproduction" to "incrementation" and "reinitiation". In particular, the prompts demonstrate how Waai shifts the "pedagogical focus" from "pedagogy" to "heutagogy", where the teacher is described as a "co-learner". This shift also involves the "locus of control" moving from teacher to student, where students are directly involved in "context shaping", thus "reconceptualising the role of the student". The result is that there is "community building" where the mobile device becomes a "collaborative tool" evidenced by "active participation" within a community, for instance, like Reddit, as a student quipped during a focus group discussion.

By ensuring that the system prompt in the backend prompting process in Waai avoids assessment or evaluation as the goal of the output,

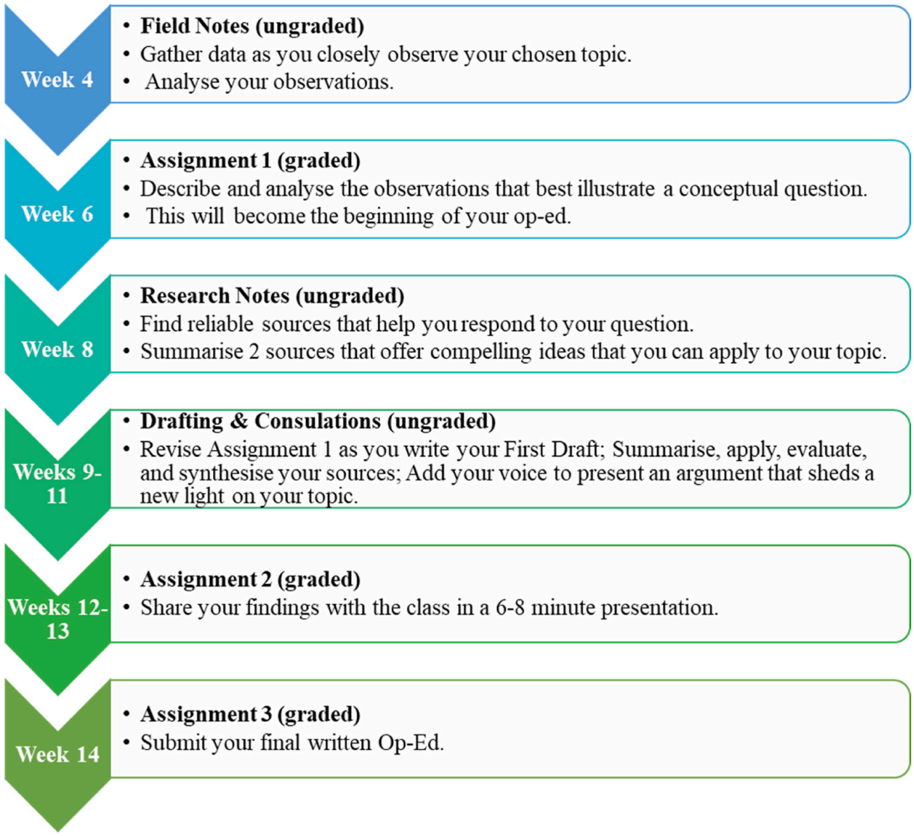

Fig. 4. Sequence of course assignments.

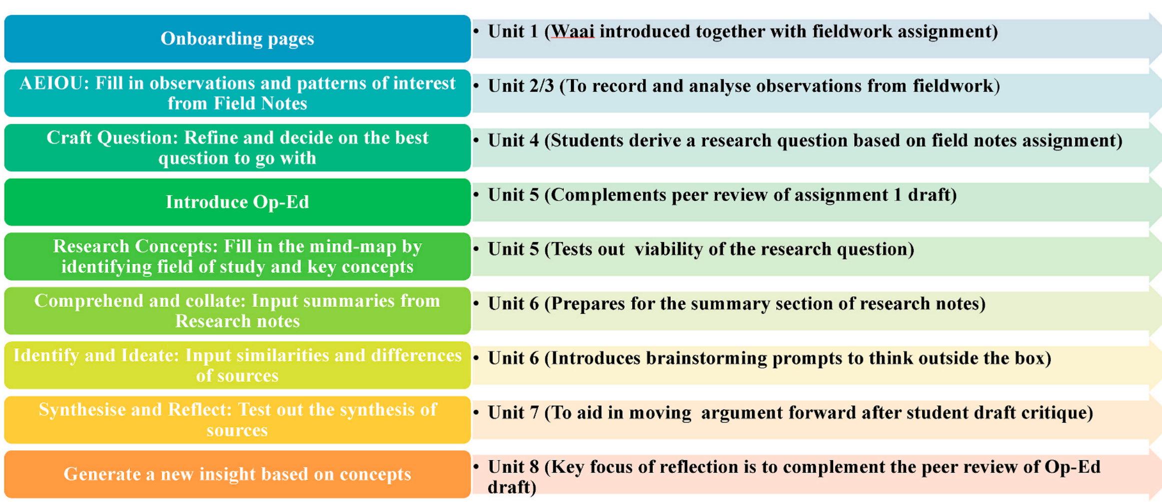

Fig. 5. Suggested Waai infusion into curriculum.

based on *the DTMLframework mapped to the Whakapiri-Whakamarama-Whakamana framework* (Cochrane et al., 2017*)*, Waai may be reconceptualising the role of the teacher through the prompt design alone.

The prompt options include prompts from the Powers of Ten, a popular design thinking brainstorming tool. The following table (see Table 2) categorises the prompts across the pages in Waai according to the New Zealand indigenous engagement framework adopted by Cochrane et al. (2017).

The column on the left reflects the non-linear 'hamburger menu' that allows students to travel to and fro in the process of ideation. The pages are, thus, editable. Right now, the prompts are separated according to the different stages of the design thinking process. According to the column on the right, the design thinking elements assigned to the different prompts suggest that Waai moves students towards 'andragogy' or 'enlightenment'. In this dimension, students are learning to 'become' without yet becoming part of a broader community that they could champion or empower yet. The 'locus of control' has shifted to the student, but the metacognitive dimension needs to be teased out, and thus, the 'incrementation' of creativity remains subjective as such. Nonetheless, students are involved in 'process negotiation'. Despite the output coming from ChatGPT-4, the content is being actively negotiated by the students themselves. This suggests that the path forward for the

Table 2
Categorisation of Prompts for Brainstorming using the Whakapiri-Whakamarama-Whakamana framework (Cochrane et al., 2017).

Pages in Waai	Prompt options for users fed to the ChatGPT-3.5 Backend Model	Using a New Zealand indigenous engagement framework to enhance mobile learning
Empathise	Help me uncover abstract concepts from my AEIOU observations	**Whakapiri *(Engagement)* Pedagogy**
	Help me uncover useful definitions of concepts relating to my AEIOU observations	**Whakamarama *(Enlightenment)* Andragogy**
	Identify any patterns or trends in my observations that I might have missed.	**Whakamarama *(Enlightenment)* Andragogy**
	Review how I can better analyse the environment and objects in my observations.	**Whakamarama *(Enlightenment)* Andragogy**
Introduce Op-Ed	Please provide insights about the vividness of my descriptions."	**Whakapiri *(Engagement)* Pedagogy**
	Suggest phrases to engage my reader's senses	**Whakamarama *(Enlightenment)* Andragogy**
	Reflect on the depth of my analysis. Have I analyzed a seemingly small detail that could be significant from my observations?	**Whakamarama *(Enlightenment)* Andragogy**
	Reflect on the depth of my analysis. Have I analyzed a pattern or a break in pattern in my observations?	**Whakamarama *(Enlightenment)* Andragogy**
Research Concepts	Recommend topics that are based on my research question, insights and interests.	**Whakapiri *(Engagement)* Pedagogy**
Identify and Ideate	What would the world be like if my topic did not exist? How would a time traveller perceive my topic? How would a Martian unfamiliar with Earth perceive my topic? How would a scientist interpret my topic? How would a social scientist understand my topic? How would an artist depict my topic? How would a creative writer narrate about my topic? How would a philosopher discuss my topic? How would a mathematician analyse my topic?	**Whakamarama *(Enlightenment)* Andragogy**
Synthesise and Reflect	How could I consider the broader societal or global significance of the topic? Why would this topic matter to someone else?"	**Whakamarama *(Enlightenment)* Andragogy**
	Does my writing demonstrate personal or emotional connections?"	**Whakamarama *(Enlightenment)* Andragogy**
	How do I include original insights in my writing?	**Whakamarama *(Enlightenment)* Andragogy**
Conclude and Refine	Review the concepts introduced in the Op-Ed. Review a key phrase or aspect of the topic in the Op-Ed. Review the last paragraph (conclusion) in relation to new insights introduced in the Op-Ed	**Whakamarama *(Enlightenment)* Andragogy**

app is to incorporate 'community building' by recreating Waai as a 'collaborative tool', in order for knowledge acquisition to move towards knowledge creation.

1.2.3. *Controlled prompt engineering*

Waai is not so much a scaffolding tool but a "learning scaffold" that targets the needs of first-year students (Lahza et al., 2025, 16) through the use of pre-designed prompts which aid in the initial stages of the brainstorming process (see Fig. 6). Rather than replacing the need to struggle through the learning process that involves adopting concepts, due to the qualitative nature of the output, we focused on keeping the output descriptive rather than evaluative. The generated feedback was framed so that users have to fill in the gaps. The way the output is presented indirectly encourages self-regulation (Zimmerman, 1989) through a knowledge gap that is inherent in initial stages of conceptual development. Moreover, the disjunction between the genre of the feedback output (descriptive) and the desired input (argumentative) controls the degree of plagiarism that might otherwise be risked in the context of GenAI. In other words, the Nudgy stops short of doing the thinking for students. However, this also requires a flexibility and openness to the suggestions in order for users to act on them in the sequence they are presented from the front-end of the app.

The GenAI output is also controlled from the backend through the use of a system prompt (see Fig. 7).

You are an experienced AI writing assistant. Your role is to provide feedback and insights on the given input. The input consists of the user data and a prompt to analyse this user data and the user has requested evaluation, identification of gaps, and suggestions for improvement. You should thoroughly analyse the provided observations and provide thoughtful and constructive feedback based on the questions posed. Your responses should help the user gain a deeper understanding. Please provide comprehensive and informative answers to assist the user effectively. Do not be purely evaluative. Refer to the user in the second person.

The chatbot's feedback was designed to address the user in the second person. In this way, we anticipated the need for self-regulation and deprioritised the need for evaluation by generating individual ideas with their own semantic and connotational capacity that encourage more relational uses of the app and asking for greater clarification in order to come unstuck in the process of brainstorming. Being an evaluative task that should not be machine-dependent (Swiecki et al., 2023), breaking up the learning process with different prompts introduced on each page, ensures randomisation in the chatbot despite the data being systematically accumulated in the built-in chatbot. Randomisation works in our favour to encourage objectivity but also to stretch the feedback output beyond predictions based on average responses in a saturated universe to formulate something new.

1.2.4. *Testing of pre-designed prompt templates*

In addition to the system prompt, backend prompt templates control the idea generation process, partly achieved through isolation of distinct parts of the brainstorming process, by defining what concepts and abstract ideas are, as well as forms of language like phrases. The following offers a sample of Nudgy AI prompt templates available in relation to the first graded assignment students submit for the module (see Fig. 8).

Based on internal testing, prompts selected based on 'concepts' and 'patterns' elicited useful, targeted feedback. Using writing input on a description of a *place*, a list of suggestions with possible connections to be explored further is generated, as demonstrated by the following Chatbot output (see Fig. 9).

Additionally, we observed that the quantity of input matters in judging the effectiveness of some of the pre-designed prompt templates. However, brevity of feedback output or fewer insights might not necessarily mean that the feedback is not helpful, as the following output suggests, (see Fig. 10) once the ideas of the user have solidified into paragraphs.

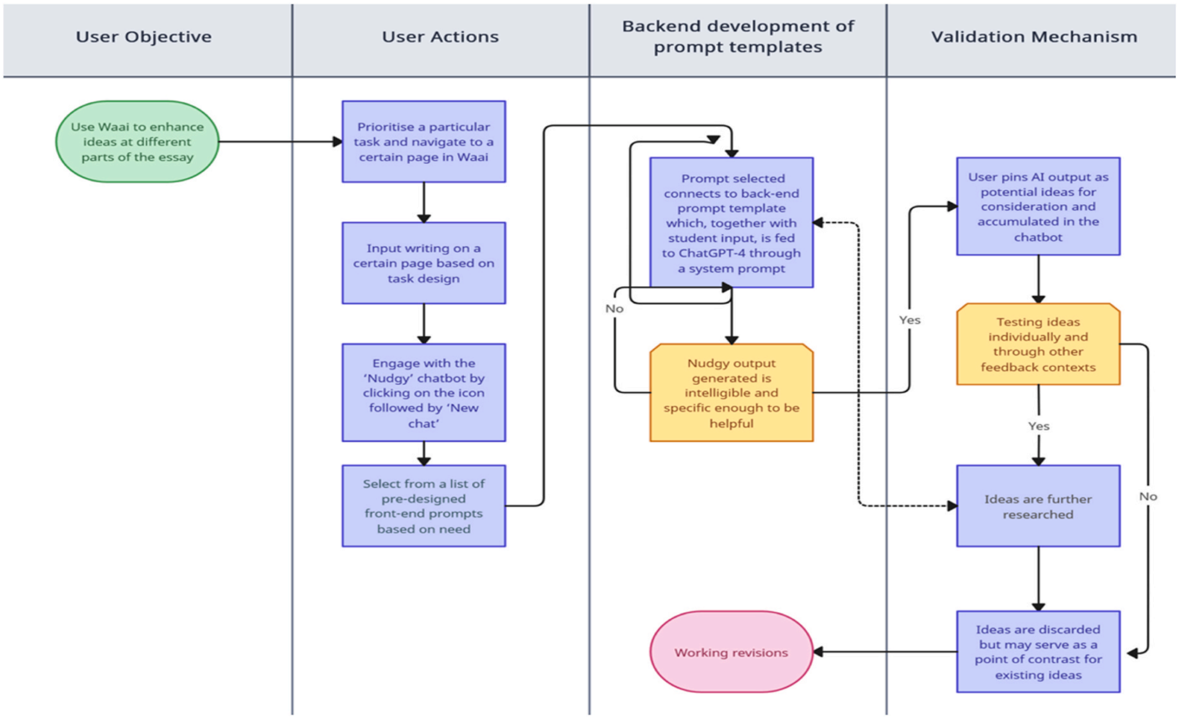

Fig. 6. Waai meta-prompt design based on an uni-directional chatbot process.

You are an experienced AI writing assistant. Your role is to provide feedback and insights on the given input. The input consists of the user data and a prompt to analyse this user data and the user has requested evaluation, identification of gaps, and suggestions for improvement. You should thoroughly analyse the provided observations and provide thoughtful and constructive feedback based on the questions posed. Your responses should help the user gain a deeper understanding. Please provide comprehensive and informative answers to assist the user effectively. Do not be purely evaluative. Refer to the user in the second person.

Fig. 7. Waai system prompt.

Prompts for "Introduce Op-Ed" Page	
Frontend prompts	Backend prompts
"Help me uncover abstract concepts from my AEIOU observations."	"Abstract ideas are *ways of thinking based on general ideas rather than on real things and events.* Your goal is to review the AEIOU observations provided, understand their context as a writer, and offer three possible abstract concepts to analyse."
"Help me uncover useful definitions of concepts relating to my AEIOU observations."	"Definitions are statements that express the essential nature of something. Your goal is to review the AEIOU observations provided, understand their context as a writer, and offer three possible definitions to analyse."
"Identify any patterns or trends in my observations that I might have missed."	"A trend is something that happens for a reason, while a pattern is the observation of things happening in a specific sequence that is predictable in some manner. Your goal is to review the AEIOU observations provided, understand their context as a writer, and offer three possible patterns or trends to analyse."
"Give me phrases that can emphasise the five senses from the AEIOU observations."	"Phrases are a small group of words standing together as a conceptual unit, typically forming a component of a clause. Your goal is to review the AEIOU observations provided, understand their context as a writer, and offer three possible phrases to analyse."

Fig. 8. Pre-designed Prompt Templates – An example.

While succinct, the above example offers some logical feedback. In being more of a general 'nudge' rather than offering specific language suggestions, this is an example of what the teachers behind the app hoped Waai could do for its users. Nonetheless, depending on the user's motivation for using Waai, the effectiveness of this output could be subjectively perceived.

To summarize, in place of a deterministic model for local verification of the pre-designed prompts, we focused on reverse engineering what we hoped the Nudgy output could achieve – creating pre-designed prompt templates based on key terms in the course book. The different stages in Waai correlate with the process of student inquiry, which begins with fieldwork observations, leads to an open-ended, conceptual research

Question prompt: Help me uncover abstract concepts from my AEIOU observations.

Response: I see you have rich observations from your nature moment. Your impressions, like the contrast between the natural and constructed environment, the silence amidst city noise, and the feeling of solitude, provide a solid foundation for abstract ideas. To offer a list of potential concepts for analysis: 1. Contrast between Man-made and Natural Aesthetics: How does the human-made environment contribute to or take away from the aesthetics of nature? 2. Solitude and Reflection in Urban Spaces: How does the solitude and privacy constant in the city lead to reflection and introspection? 3. The Paradox of Silence and Noise: At what point and why does noise turn to silence on a psychosensory level? 4. The idea of Nature as an Escape: How does nature serve as an escape mechanism within an urban setting? What triggers are present? Use these ideas for your analysis/ Great start!

Fig. 9. Nudgy output example – "Uncover abstract concepts".

Question prompt: Please provide insights about the vividness of my descriptions.

Response: Your descriptions are indeed vivid, especially when you narrate the unexpected wilderness and your emotional connection to it. Some parts, however, could use more specificity for clarity, such as your 'pressing inner reality'. Detailing this could enhance readers' understanding of your feelings. Similarly, when you discuss the transformative power of writing, concrete examples could substantiate your perspective. Associating specific excerpts from your mentioned sources would solidify your statements further.

Fig. 10. Nudgy output example – "Insights about vividness of description".

question, and proceeds to brainstorming for key terms to research to respond to the research question. As shown in the examples, the output elicits information like concepts and disciplines of thought essential for developing original schemas of thinking, which require but also shape, the user's ongoing research. Waai relies on the student user to research more into each suggestion before implementation. Moreover, the implementation involves several stages including literal comprehension of AI output, defining, if necessary, the key terms, synthesizing AI output to develop students' own ideas, and creating an original argument or new insight. Thus, in developing a fully-fledged app, we felt that the testing of the accuracy of the output is not as essential as aligning the app with the aims of brainstorming. Instead, we relied on testing, refining, and modification, paired with student feedback through surveys and focus group discussions, as well as concurrent updates to ChatGPT to ensure the backend process runs smoothly. Essentially, we approached GenAI as a 'black box' of creativity.

2. Methodology

After acquiring IRB approval from the PI's university, 130 Students from the PI's classes were recruited to test the app over two semesters (August 2023–May 2024). Of those 130 students, 31 students volunteered to take a survey, and of those 31 students, 11 volunteered to take part in focus group discussions. To acquire a finer grain of detail for the 11 FGD students, we also examined the interactions these 11 students had with Waai, to confirm their self-reported perceptions. Students received small tokens of appreciation for their participation. They participated with the knowledge that we would remove any identifying information aside from the students' disciplines during the research.

2.1. Student participants

All participants had spent a semester using the Waai app while taking *Communication and Inquiry in an Interdisciplinary World*, the course for which the app was designed. Surveyed participants were from a range of disciplines, though the dominant discipline represented was engineering. Focus group discussion (FGD) students also came from a range of disciplines, including business, sciences, engineering, social sciences, and communication studies.

2.2. Surveys

Qualtrics was used to conduct a survey at the end of each of the two semesters. 31 students volunteered to take the survey and were assured anonymity. A mix of close-ended and open-ended questions covered different elements of the CC0001 writing process and the different parts of Waai that target this process. (see Appendix A). Students were also asked about ease of use and what they desired in terms of the user interface and visual design of the app. Students were informed it might take maximum 10 min of their time. In this survey, GenAI tools are defined as tools that help in the process of brainstorming and/or writing, accessible on a computer or mobile device that contribute to a defined learning purpose related to brainstorming and/or writing.

2.3. Focus group discussions

Students who did surveys were recruited for focus group discussions on a volunteer basis after final marks were submitted. The in-depth FGDs followed a semi-structured format. Students were asked about their experiences with Waai, specifically their interactions with the Nudgy GenAI chatbot. Particular attention was paid to their interactions with the Nudgy and how it might have helped them brainstorm and form arguments (see Appendix B).

Despite the small pool of students involved in face-to-face interviews, the data complemented the survey questions as students were prompted to offer elaborations on their use of the app. Despite only having about 15 % of the survey pool interviewed, generalization is possible because of the randomisation of students who signed up in groups for the FGDs, who represented a diverse variety of faculties. We also studied students over two semesters, to discern if the initial success of the app when first piloted could be the result of the 'anticipation effect,' a concern with objectivity especially in the context of educational technology.

2.4. Data analysis

Transcripts of FGDs and survey data were analyzed using a 'discovery' approach (Leong, 2013) where recurring themes in the transcripts were inductively coded. The codes and coding process were checked and reviewed by each member of the research team independently. To supplement this data, we generated a spreadsheet by exporting the data from the app, which distinguished individual students into rows, where the different columns indicate user input as well as output from the chatbot as part of the data collection for the app. With this data, we also reviewed students' final assignment submissions and teacher feedback when relevant.

3. Results

For the results of the survey, it was fairly unanimous that students felt that Waai was helpful in terms of supporting them in their use of concepts though the ability to drill down to specific concepts should be improved. It was felt that a balance between a highly structured design and unstructured access would be necessary for the effectiveness of the asynchronous engagements hoped for within the app (see Fig. 1).

As Chart 1 shows in more detail, survey participants found the AI chatbot of Waai, Nudgy, most helpful for analysing their observations, suggesting topics based on the students' research questions, finding concepts to research, and brainstorming/ideation. They indicated that the AI feature was least helpful for synthesizing their sources and refining their conclusions. This reflects the designers' intentions when making Waai, as we wanted the app to help students with brainstorming rather than editing their work. We also purposely avoided allowing the AI to help students with any of the communication skills that were learning objectives of the course, such as synthesizing sources.

Below, we have divided our findings into positive and negative experiences with the GenAI Nudgy chatbot of Waai. Overall, students were able to benefit from the Nudgy in 5 distinct ways: 1) its pre-engineered prompts, which were tailored to the course assignment rubrics; 2) its tendency to recommend topics to research rather than give students answers; 3) its suggested research topics, which helped students to consider different perspectives on their topics; 4) how it modelled ways to ideate new insights; and 5) its constant availability. Students, however, expressed reservations about the Nudgy, particularly in terms of: 1) the limitations of pre-engineered prompts within the app; 2) difficulty in discerning the most relevant of the Nudgy feedback; 3) mistrust in GenAI and Aigiarism; and 4) a recognition of the limitations of GenAI in supporting argumentative writing.

3.1. Positive experiences with the GenAI chatbot nudgy

1. Student appreciation for pre-engineered prompts

In the survey, students were evenly split when asked if they would refer Waai or ChatGPT to a friend. For students who would prefer Waai, they valued how the prompts were designed to complement the course curriculum.

In the student survey, the second feature that students found most helpful in Waai was the pre-designed prompts that were fed into the Nudgy (average 3.93). In the focus group discussions, students such as P5 from environmental sciences, appreciated how the AI questions were tailored to the class's learning objectives. Likewise, P3, a communications student remarked:

> "for an AI, they have like such a huge database that we can leverage on. So I think that it was really helpful in terms of giving prompts. And actually, I think that I agree that, as compared to other AIs out there like ChatGPT, this is a lot more specific and tailored to our experience. So I would think that, like they are able to give more concrete information as compared to the AI that's out there." (P3)

2. Nudgy generating ideas, not giving answers

Students repeatedly mentioned that they valued being able to make choices as writers and have agency over their work. Therefore, they appreciated how the Nudgy chatbot would brainstorm along with them, providing lists of possible ideas, without doing the writing for them.

For example, P2 from biological sciences liked how "instead of just spitting out answers" the AI chatbot in Waai allowed for "freedom" and "creativity." It was important for this student that her writing was her own work: "it really questioned us [… and] really got us to think about our own work, and like we still feel that, you know, it's our own work." P5 similarly noted that he could have agency as a writer in "vetting" the ideas from the Nudgy.

In the survey, one student explained that they wanted to avoid ChatGPT because it can "prescribe answers for you rather than encouraging thinking." Another student echoed this in more detail in the survey:

> "I think I would recommend Waai because personally […] I think Waai is more customised to fit [the module] and I also think that as compared to ChatGPT I like how Waai does more of probing than giving an answer as I feel that ChatGPT gives answers which may in turn restrict our ways of thinking as we may think the idea they give is good and instead be harder to think of alternatives." (survey)

Interestingly, this student hits upon one of the team's main concerns: that in prescribing answers for students, AI could subsequently limit a student's own ideation process.

Several students explained that they appreciated that the chatbot prodded them to brainstorm on ideas, without explicitly prescribing what to think. P4, a business student, noted that the AI chatbot can "generate ideas to kind of like trigger, like your own thinking from it" rather than tell you what to write. P3 echoed this sentiment that the Nudgy acted more as a "trigger" than prescribing ideas, noting the chatbot was true to its name in "nudging" the user: "It doesn't tell you what to do, but it reminds you that you have to go and do something." P10 affirmed this positive experience with the Nudgy: "it'll just prompt you. It'll probe you in the right direction, but not give you the entire answer itself, which is really necessary to get you thinking in the right direction."

When reviewing the interactions students had with Nudgy, we could confirm that the chatbot was able to give recommendations for topics to explore, without providing answers for students. For example, as P7 was exploring how trust is formed in online business transactions, the Nudgy suggested that the student reconsider the importance of contractual agreements and business relationships (see Fig. 11).

In general, the student was able to identify different characteristics of

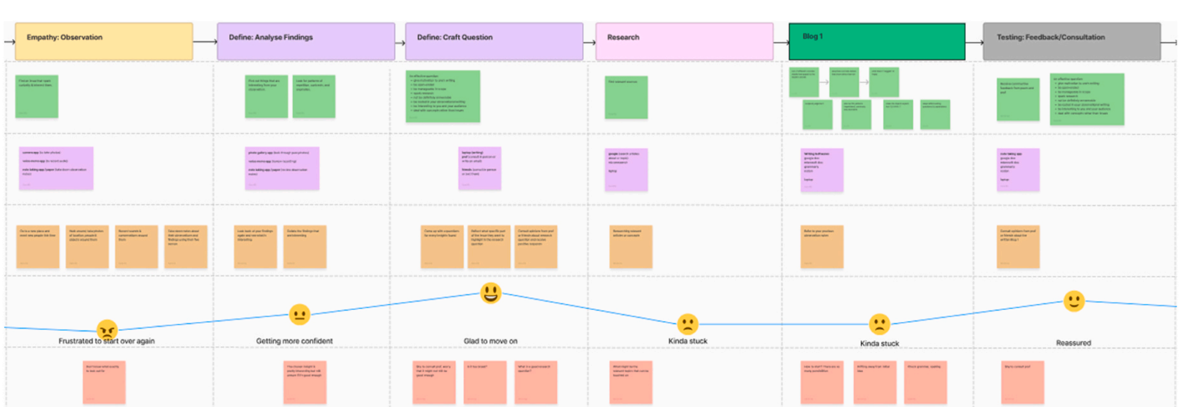

Fig. 1. Initial ideation of user journey experience using emoticons.

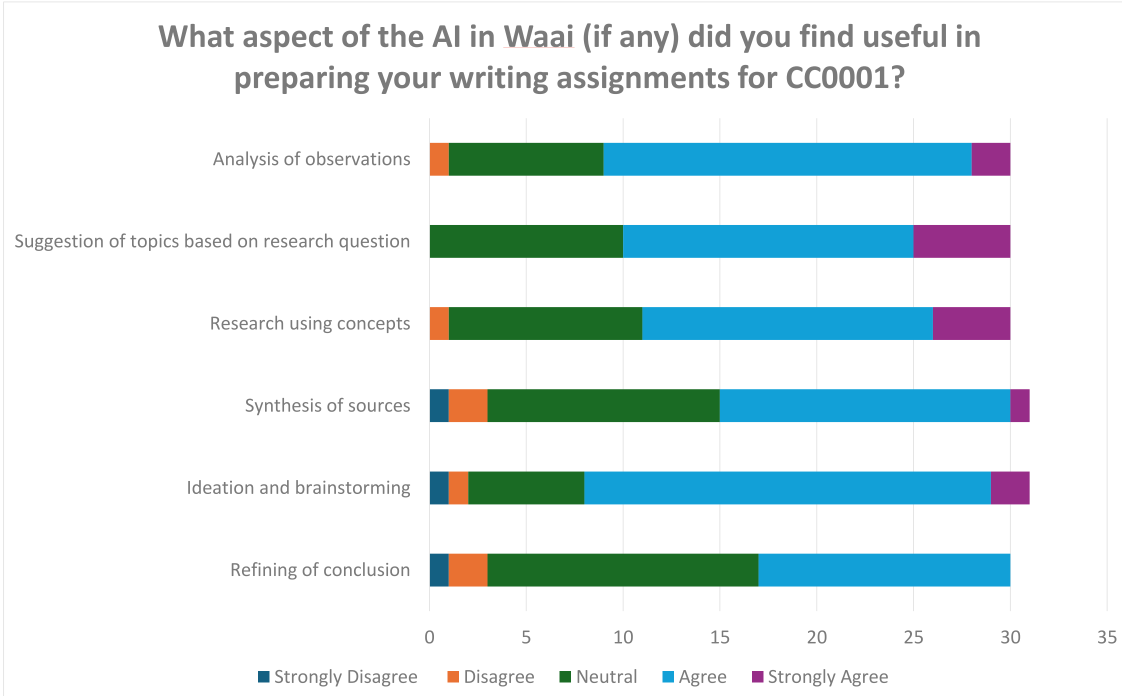

Chart 1. Aspects of the Waai app students found helpful.

Question prompt: Review a key phrase or aspect of the topic in the Op-Ed.

Response: The Op-Ed discusses the limitations and importance of trust in professional relationships. It highlights the need for vulnerability and mutual risk-taking for trust to develop. The author also mentions the significance of partner selection based on shared values. However, the argument against using contracts and relying on mutual obligations seems short-sighted, as self-interest and the need for protection can outweigh trust. Overall, the Op-Ed provides insightful perspectives but should consider the complexities of business relationships and the role of legal agreements.

Fig. 11. Nudgy output for P7 from the 'conclude and refine' page.

business relationships but could have progressed to consider the ambiguities of the topic beyond the ideas discovered based on the two sources the student cited. Interestingly, the feedback given by Nudgy on the student's conclusion was similar to the instructor's feedback (see Fig. 12).

3. Nudgy assisting students to get different perspectives on their topics
   Students repeatedly discussed how they benefitted from the Nudgy's ability to "recommend topics" (P8). P7 from accounting remarked that the Nudgy "help[ed] us to think, like help[ed] us to delve deeper into the topic that we are already writing on." Likewise, P1 from engineering noted that the Nudgy helped to "expand your idea."

P8 from accounting also utilized what she called some of the "more creative" prompts which asked questions such as what a Martian or people with different disciplinary views would think about her topic. She said that these responses from Nudgy helped her to come to a more "interesting insight into, like a different point of view of my" topic. These suggestions from the Nudgy gave her new concepts to research, and, as P8 described, "different topics that I wouldn't have thought of before."

Referencing the prompts that asked Nudgy to present differently disciplinary views of a student's topic, or to imagine what a Martian might say about the student's topic, P9 appreciated the Nudgy response to such questions: "it gives, like various perspectives, which is quite helpful in helping us like, think of, maybe another way to approach our research question. Yeah, I think that's great."

Aside from giving students different perspectives on their topics, the Nudgy also helped some students get a "clearer direction" (P3) or "path" (P5) for their final argument. P10 suggested that the Nudgy responses helped her understand how a reader might interpret her ideas:

[Compared to the individual source analysis], the conversation [between the sources] more accurately suggests that the sources are actually complementary. However, the conversation would be mostly a summary of the sources.

You would imply, and could have made more explicit, that manufacturer-retailer relationships are more dependable and predictable than actual relationships involving a hierarchy.

The point about sanity may be an important one – but what is its connection/ relationship to a sense of obligation in relationships?

Fig. 12. Excerpt of instructor feedback for P7's final written assignment.

"I think I used the Nudgy feature a lot to uncover any implicit concepts within my observations or sources. […] there were some underlying assumptions that I made in my Op-Ed which needed further unpacking. So this further exploration and probing wherever needed was suggested to me by nudgy, which helped me a lot to actually make sure that the reader understands where I'm getting at right and despite there were some occasional delays in response times, right, but I think the nudgy insights were consistently profound. Right? I think they were really good and suggesting me to explore different ways that the reader can interpret what I'm trying to say." (P10)

Overall, we see that the output of the Nudgy (based on pre-engineered prompts) allowed students to gain new perspectives on their topics and develop their arguments.

4. Nudgy modelling ideation

P5 clarified that he struggled to find concepts and so found that Nudgy not only provided ideas for concepts to research, but also modelled for him how one might brainstorm for ideas:

"I really struggled for this kind of like ideation and finding concepts that are related. So it was really good to- cause the AI also explained why the contents might be relevant. So it helped me get an example of the thought process that I can use to find other ideas that might be relevant to my own." (P5)

P4 also noted that the Nudgy "actually leads me to somewhere that I didn't know that I could […]. So I would say it was actually quite helpful, as in kind of like, I form my ideas, generate ideas, and it brings me along to my final research questions." By "leading" the student to ideas he did not realize could be related to his topic, the Nudgy modelled for the student how concepts can be associated and connected.

Likewise, P11 implied this when she noted that "in terms of venturing out [of] the comfort zone, I think AI actually enables that." In other words, the AI output showed the student how to consider topics to research that were beyond what the student would normally consider.

5. Nudgy available 24/7, alleviating reliance on teacher

P2 liked how, unlike with a professor, you could ask Waai questions 24 h a day, even when working late at night. A few students also noted that without the Waai app, they would have felt more reliant on the teacher of the class: "if not for the app, I think that I would be more reliant on Prof […] to like, help me in like trying to fix a specific route" (P3); "I probably would have been more reliant on Prof […] cause I would need to get all that ideas or prompts from somewhere else" (P5).

3.2. Negative experiences with the GenAI chatbot nudgy

1. Frustration with pre-engineered prompts

In the survey, students were evenly split when asked if they would refer Waai or ChatGPT to a friend. For the students who would refer ChatGPT, they tended to present the reason that it is more "user friendly." This was mainly because students had to input all their findings and writing progress into the app in order for the Nudgy to function. Some students also preferred to be able to design their own prompts for the chatbot, rather than always follow the pre-engineered prompts.

Thus, there was ambivalence among students on two main features: 1) some appreciated how the Nudgy made use of the writing they fed into the app, while others were frustrated that the app required them to record their entire writing process; 2) some appreciated the pre-engineered prompts that were tailored to the course, while others wanted the freedom to interact directly (design their own prompts) with the AI chatbot.

Relatedly, P11 suggests that students would like to see the chatbot perform an expansion and contraction function to give them feedback on how they are actually applying the output-feedback to their writing – is the feedback leading them astray or helping them further concentrate their ideas? The desire to have this mechanism in the chatbot reflects the prompting process offered by GenAI tools like GPT-3.5 and GPT-4. However, the desire is that this mechanism is "built-in" for student users. As P11 explained, she preferred "unsolicited prompts" because "as students, we tend to be very doubting of what we write. Because we don't know if this is right or this is wrong." She further clarified that although she understood that the course aimed to guide students "to use ideas creatively, […] to expand on your ideas instead of restricting them" that his came with the danger of students not realising when they were going on tangents. Therefore, she expressed a need for the Nudgy to provide unsolicited guardrails for writers: "as you move down these steps, maybe if the Nudgy could prompt like when things go a bit out of line, or when things are a bit messy, how to improve" (P11).

2. Nudgy generating ideas to research but difficult to "drill down"

One student explained that while the Nudgy helped her to find related research concepts when brainstorming ("branching out," P11), it could not aid her in drawing her concepts together or in delving deeply into one concept:

"I just felt like there was quite a bit of difficulty drilling down the concepts. Yeah, because it gives very, very generalized examples, and very generalized prompts as to how to continue. Because it makes you question yourself. But I mean, I mean, it's good to question yourself. But then, when it comes to actually like finding ways to synthesize, maybe 2 different concepts, it's a bit difficult to use it – I mean, use Nudgy to help" (P11).

As P11 summed up, the Nudgy was best at "expanding on ideas rather than concentrating them" (P11).

Looking at student interactions with Nudgy, the following feedback output for P8, on the Identify and Ideate page, lists concepts in no particular order (see Fig. 13).

For the student to make use of the Nudgy output, macro concepts like 'existentialism' and 'scientific determinism' would need to be differentiated by the user from the other concepts in order to be useful, rather than simply listed. 'Human curiosity and understanding' should also not be lumped together. In this case, 'understanding' refers to the 'scientific understanding of the universe' which, unless the student read into it, could elude them. Moreover, the 'boundaries of human perception' would more specifically point to ideas and concepts from psychology while human curiosity might be seen to be a more philosophical standpoint. Finally, concepts like rebellion and acceptance imply a binary relationship that has implications for theories like free will, whose association is not consciously made by the GenAI output. Closely studying the output, thus, suggests that a student may feel unconfident to apply any of this feedback, in this case concepts, in their work.

P8's research question in the end did not include concepts such as the 'absurd' even though the student was able to define this concept appropriately in her assignment. Thus, it might be seen that only listing and defining these concepts may not necessarily help a student make the 'leap' to the actual application of ideas in the written essay. As seen in the following excerpt from P8, the student was able to identify useful ideas, and re-present them in the writing, but was unable to dig deeper into them to explore the implications (see Fig. 14). The writing would become superficial, rephrasing ideas, instead of going the extra step towards analysis. In this sense, the act of revision should be able to suggest that a ready-made or acquired structure and order are secondary to ideation and the trial-and-error process involved in it.

Given that part of the objective of the main assignment is to connect two sources, the actual application of concepts requires additional support, the absence of which may explain the inadequacy of the ideation process for this student.

Question prompt: How would a philosopher discuss my topic?

Response: When delving into philosophical discussions about your topic, some main themes or debates that might arise are:

1. Existentialism vs. Scientific Determinism: The tension between contemplating the purpose and meaning of life (as explored by Camus) and the scientific understanding of the universe (as discussed by Deutsch), raises questions about free will, determinism, and the role of science in shaping our understanding of existence.

2. Absurdity and Meaning: The concept of the absurd, as presented by Camus, challenges our quest for meaning in a seemingly indifferent universe. This theme prompts discussions about the nature of meaning, how it can be found or created, and the consequences of living in a world seemingly devoid of inherent purpose.

3. Human Curiosity and Understanding: Deutsch's exploration of the multiverse invites philosophical discussions about human curiosity, our desire to understand the universe, and the limits of human knowledge. This theme raises questions about the nature of scientific inquiry, the boundaries of human perception, and the relationship between scientific and philosophical investigations.

4. Rebellion and Acceptance: The contrasting approaches of Camus and Deutsch offer perspectives on addressing existential dilemmas. These perspectives prompt debates about the merits of rebellion against the absurd, seeking personal meaning through embracing life's challenges, versus accepting the limitations of our existence and finding solace in scientific understanding.

Consider exploring these themes and angles to delve deeper into the philosophical discussions surrounding your topic.

Fig. 13. Nudgy output for P8 from the identify and ideate page.

Despite the seeming pessimism, Camus argues that accepting life's absurdity is a call to action against the meaningless rather than a cause for despair. He suggests that discovering personal meaning, joy, and fulfillment in the mundane components of life—even in the face of innate absurdity, is what characterizes human resilience and freedom (Camus, 1942).

SC ?

At first glance, these works might seem disconnected—Deutsch's scientific exploration seemingly distant from Camus' existential ponderings. However, in his reflections on the absurd, Camus grapples with the pointlessness of human endeavors, reflecting the monotonous nature of Sisyphus's work—a view that is also reflected in Deutsch's discussion of the multiverse, in which parallel universes appear to replicate indefinitely.

What are the implications of this point of similarity between the sources?

Nice.

Moreover, both works explore the fundamental questions of existence. Deutsch subtly tackles the core question of human curiosity—our thirst for understanding the universe and our place within it (Deutsch, 1998). Camus, on the other hand, confronts the discomforting reality of an indifferent cosmos and the human struggle to find purpose, resonating with the unease that arises when faced with the absurd (Camus, 1942).

However, the divergence between these works lies in their responses to these existential dilemmas. Rooted in the domain of science, Deutsch charts a course forward through exploration and inquiry. He presents the multiverse idea as an open invitation to explore, comprehend, and possibly even find comfort in the universe's complexities rather than as a declaration of hopelessness (Deutsch, 1998). Camus, in contrast, emphasizes the importance of rebellion against the absurd. He urges us to embrace life's obstacles as a means of uncovering our own unique truths and ideals in the face of the meaningless in order to find personal meaning and purpose (Camus, 1942).

SC Rephrase

SC Unclear transition

SC How so?

Fig. 14. Screenshot of an excerpt of P8's final essay with instructor's feedback.

3. Concerns of academic integrity and mistrust in AI

Based on surveys and focus group discussions conducted over two semesters of testing the app with the PI's classes, the team noted that students perceived limitations in the controlled GenAI environment of the app as well as concerns with Aigiarism. In particular, some students felt that the necessity to declare GenAI usage makes them more cautious about using any form of AI, while others feel like the need to declare AI usage only affirms the potential of GenAI to help students gain an advantage, where used correctly.

P3 mentioned that her sense from university policies is that students should only use GenAI to "prompt or spark inspiration." Therefore, it was helpful to have a teacher introduce the students to the Waai app and be encouraged to use GenAI in this more controlled way within the module. P5 also said that he wanted to avoid AI to avoid plagiarism; he explained: "when I use AI, it feels like it's not my own work."

Relatedly, P11 stated: "I think it [GenAI] kills the creative process in terms of allowing us to think for ourselves." Likewise, P10 argued that "currently there is this over-dependency on AI, and I think it really kills the creative process." While P10 noticed that her peers used GenAI to generate text, she felt that "ideas should be your own, [they] shouldn't come from AI" (P10). Instead, she explained that she used ChatGPT to illicit feedback on her writing (P10).

P3 noted that they were wary of "trusting" Nudgy to be "correct." Instead, she explains that she used the chatbot to get exposure to concepts she might not have thought of, which she can then "build" on:

> "It tells you that like maybe it is this, but it's not like 100 % this, so you should go and search more about it, and it gives you more of like an example of some concepts that you might not have been able to think of, and hence then you can build up upon it." (P3)

4. Nudgy limitations

A few students also noted that GenAI cannot provide you with your own opinions and judgements. For example, P9 explained that GenAI cannot convey a "personal voice." P11 elaborated on this idea, explaining:

> "AI won't be able to provide you a personal take on your own personal experience. So, I think, yeah, AI definitely didn't help with that. […] I had to draw on my life experiences and ways in which my own perspectives would have impacted the way I viewed those concepts itself." (P11)

4. Discussion

Based on this pilot study, we are convinced that when used as one element within a brainstorming process, rather than the only tool for ideation, GenAI may help motivate students to ask "so what" and even train student readiness in terms of the process of revision. By attempting to control the nature of the output, especially at the initial stages, we presented Waai as an efficient processing system which was persuasive especially to student users unfamiliar with prompt engineering. Efficiency was determined largely by the speed of processing rather than based on the effort needed eventually to apply the feedback and/or revise the writing. However, beyond that, Waai encourages and even requires rethinking the role of confusion in building cognitive capacity (Lodge et al., 2018). Below are our central conclusions.

1. While students appreciated course-specific, pre-engineered GenAI prompts that did not prescribe answers, they ultimately were ambivalent and remained reluctant to let go of the ease of having their own "conversations" with GenAI.

Unlike a Socratic questioning chatbot, as interview subjects noted, Waai does not offer answers but rather responds directly to student writing at different stages of the writing process. The following flow chart shows the different iterations of Waai. The initial objective of creating a writing app, and even a writing assistant app would either not allow us to distinguish Waai sufficiently from other available apps or offer questionable help that might replace actual student ideation. In the end, we found the conversational structure initiated by a chatbot to be the most useful feature. Despite the limitations of GenAI as a thinking tool, the chatbot feature could be relevant in more open-ended contexts of brainstorming, which also suggests different ways in which the "Nudgy" might be engaged with on the Waai platform (see Fig. 15).

The pedagogical role played by Waai includes the "supporting learning role" where students learn how to apply new concepts picked up through a conversation with the chatbot, as well as the "assisting role" where students can be reminded of the various tools of investigation like zooming in on a new idea to analyse it further. Finally, together with other brainstorming tools built into the app, the chatbot also has a "mentoring role" whose intention is to encourage recursive ideation as a metacognitive approach to the writing process (Wollny et al., 2021).

Further to the above, in considering the potential of GenAI for personalized learning (Mygland et al., 2021), instructors need to consider a human-AI symbiosis framework (Litvinova et al., 2024) that brings together input from the pre-designed prompts drawn from the curriculum material, the process that students engage with using the uni-directional chatbot, and how the output validates both the use of Waai as well as student progress in thinking.

Thus, with our pre-engineered prompts, students were unable to plagiarise the ideas by copying and pasting whole sentences into their assignment. Instead, they were engaged in a process of brainstorming necessary for content revision as opposed to superficial editing.

2. While students appreciated how a GenAI chatbot could suggest and model the exploration of topics they never considered, they remained wary of its correctness, plagiarism and losing their own voice.

The surveys and focus group discussions over two semesters revealed that students perceived limitations in the controlled GenAI environment of the app as well as concerns with Aigiarism. Yet, as the primary purpose of Waai is brainstorming through the consideration and testing of theories and concepts, the output from Waai may not have a direct impact on student written assignments. Rather, the impact depends on how students choose to use the Nudgy feedback. Thus, in addition to affording technological access, we realised that it was important to address the key underlying motivations for using GenAI in higher education, and the role of teachers in ensuring that critical thinking standards are not compromised.

In this way, GenAI could be used to emphasise the need to encourage student engagement with feedback to promote a more interactive feedback loop in the design of assignments that may allow and even require the use of GenAI. Developing GenAI literacy, as observed from the FGDs, is an ongoing and evolutionary process, which could lead to not just more enlightened use of GenAI but also a deeper understanding of the ecology or environment for developing the writing voice.

The key assumption in Table 3 is that, with critical thinking in view, encouraging a more sociocentric mindset may lead to greater development of a writing voice. Thus, how can users be encouraged to move from egocentric to sociocentric goals in both their approach to writing as a process of exploration and the use of educational technology as a personal assistant? We argue that the prompts designed play a large part in establishing power and knowledge relations in response to the use of AI in education. The prompts act as both objects to prompt thinking as well as an opportunity to question their own application of critical thinking especially around concepts.

3. While students found it useful when the GenAI chatbot offered different perspectives on their topic, they found that it could not help them go deeper into their topic.

Within Waai, the listing structure of the feedback perhaps made students feel overwhelmed and unable to choose certain ideas over others. Thus, this could affect the perception of the efficacy of the AI chatbot, which a multi-modal chatbot (in possible future iterations of the app) would be able to address. A significant reflection from students is that feedback that is not evaluative might be too 'general' and thus

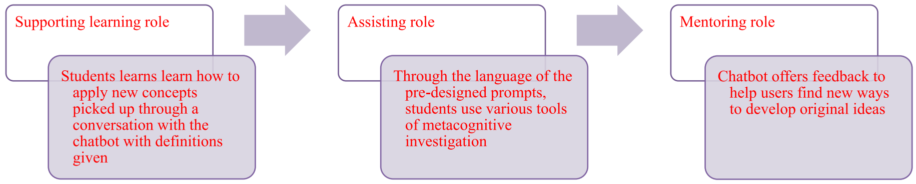

Fig. 15. Different pedagogical roles represented by Waai.

Table 3
From egocentric goals to sociocentric goals to justify the **use of GAI**.

Egocentric goals	Sociocentric goals
If GenAI can make my writing better, will I be penalised for it?	How will GenAI help me find my own voice and therefore find an audience for my work?
Can GenAI think for me so I can complete writing tasks more efficiently?	How can GenAI help me move from informational writing to more nuanced arguments that go beyond an argument as debate?
Can GenAI provide feedback for me that will make up for the lack of complexity in my ideas?	How can I use GenAI to select and integrate useful ideas that can contribute to a broader readership?
Can GenAI tell me if I should be getting an A?	How can I use GenAI to sharpen my thinking process and think critically about the concepts from the fields of inquiry I am engaging with?

prove to be more generic than useful. This is the dilemma of designing prompts that do not simply perform the thinking for students. This may come down to our perception of *what brainstorming is in relation to ideation*—beyond the context of a formal writing assignment.

The efficacy of the output was not the only measure of students' expectations of Waai, but how it brought them to a stage where they could deepen their engagement with ideas (Ananthaswamy, 2023). Thus, there were important comparisons, largely volunteered and unsolicited, made between ChatGPT and Waai in relation to their usefulness as brainstorming tools. Some factors raised included the deterministic nature of the prompts. Significantly, the accuracy of the assessment of writing indicators like fluency also came up, implying a desire that Waai might also incorporate assessment as part of its learning design. This is probably due to the simultaneous nature of the use of Waai with the submission of individual assignments.

4.1. *Pedagogical implications*

The social construction of technology as a research heuristic from cognitive science (Bijker, 2001, 2015; Pinch & Bijker, 1984) claims to support the process of ideation without being deterministic. This is essential in the field of intelligence augmentation such that theory and practice blend where GenAI is used not just to inform current practices in technology but to develop an already established pedagogical method of scaffolding.

Waai is a prime example of where the human-AI feedback loop is core to app development, as well as an important aspect of writing instruction that goes back to Socratic thinking and the importance of asking questions over finding answers. Likewise, GenAI could be essential in suspending disbelief where problems are sought rather than necessarily solutions, an important aspect of thinking outside the box needed in the leaders we hope to shape for the future.

The process of creativity requires distancing from the materialist positions that inhibit thinking outside the box. As opposed to didactic teaching or abandoning all instruction to chance, instruction encompasses carving out opportunities for ideation. The former echoes how spurious claims of scientific information which ChatGPT has been associated with have resulted in less than scientific claims or "hallucinations" where partial truths are offered. In a sense, the question of voice has always been under the spotlight where there is a tension between writing as representation and writing as nature. Is writing in the first place to report an existing idea or to provoke thinking? If the latter, how does "Waai" shape writers by claiming that it supports and even mentor student writing?

Waai might be seen as a case study belonging to the paradigm of "AI-supported, learner-as-collaborator" where individualized information is collected within the app to customise the feedback. The main issue here is to ensure that the AI-supported feedback is both adaptive and informative enough to pave the way for student writing to reflect a distinct improvement in the ideation of an exploratory argument (Ouyang & Jiao, 2021). The role of GAI remains almost paradoxical in relation to developing thinking especially where AI assistants appear to have overtaken writing especially in practical domains. However, we believe that it is necessary to study its potential for knowledge creation.

A key reservation arises in relation to the language prompts on Waai that support the writing of vivid description using the five senses as part of their initial Field Notes assignment. Although the Waai prompts operate within constraints, as seen from the system prompt, yet they also offer the fiction of aesthetic judgement though Waai is asked not to be evaluative.

Here is an example of the output when a language prompt on Waai is selected (see Fig. 16).

As a result of selecting the prompt "Suggest phrases to engage my reader's senses", some possible phrases were generated as examples of descriptive or sensory description. While the phrases do combine different elements of 'nature', how they contribute to new understanding about what 'nature' represents and how it might be particular to the writing input shared, may be unclear. The purpose of these phrases is to lead to a concept to analyse. These words are seen to be standalone phrases without a sense that there are multiple layers of differing complexity in reading them – e.g. Are these images based on a point of similarity or point of difference, a form of comparative analysis that leads to new thinking, and what is the effect of that if introduced in the actual writing? Moreover, if all the data that the AI is trained on generates clichés instead, these could be hinderances to good thinking – unless of course the point is to use a cliché and demonstrate its affective impact. While the writing input is original and has some basis in the writer's own experiences, having been derived from actual fieldwork, the ideas in the output are derived in isolation from that reality. ChatGPT would not know the effect of using the metaphor of a "symphony" in the way a writer would if they looked up the origins of the concept and the tensions that it evokes in relation to the evolution of classical music.

The applicability of these phrases to the actual writing needs further thought and evaluation from students, preferably in a collaborative environment. In other words, for an honest account of the uses of GenAI for the purposes of "learning", the myth of intrinsic technological effects (Lodge, Thompson, & Corrin, 2023) needs to be exposed. This is the fallacy of conflating automatic GenAI output with the dream of "automatic language production" (Tenen, 2024). Tenen (2024) suggests there needs to be "a comprehensive middle language" to make sense of the disorder of human writing. However, because a truly automated language for original discourse is not forthcoming, what is at stake is "novel discourse" which Tenen suggests puts "the logic of composition" that goes beyond basic learning of information (46). This is where we could

Question prompt: Suggest phrases to engage my reader's senses

Response: Your introduction is wonderfully descriptive, and it's easy to see your enthusiasm for the topic. However, there's room for more detailed sensory language to further draw readers in. For instance, you could describe the colors, sounds, and smells around the spot of wilderness to titillate the senses. Try adding phrases like: - "The vibrancy of the flora, with the verdant green leaves contrasting against the starkly bright wildflowers" - "The symphony of rustling leaves and birdsong filling the air" – "The fresh, earthy fragrance characterizing the wilderness". A balance of narrative and sensory language will make your piece all the more enticing!

Fig. 16. Nudgy output example – 'Suggest phrases'.

find a reason to write for the purpose of human advancement, as Tenen suggests.

For this reason, a customized chatbot is better than free prompting on Chatbot, anytime. The challenge of avoiding a focus on right and wrong answers and therefore an open research question suggests there might be a right or wrong way of addressing GenAI output especially in the context of "learning" — where the pedagogical aims of Waai remain under the expertise of the human creator of the prompts. This began with limiting what a concept means, which in turn limited the output from ChatGPT-4 to words or short phrases. Thus, the "middle language" on Waai is implied behind the way the output was designed to avoid obvious links beyond that of individual concepts and phrases.

Likewise, a 'barrier to entry' then, which may be initiated through evaluative activities that actively include collaborative activities from a class poll to structured peer review, may be essential to strengthening, to use the same example, the usability of these phrases as points of inspiration for original writing. While the procedure of sharing an input resulting in an immediate output from the AI based on a backend process has the potential to cause students to think differently about their writing, the personalization of feedback is just the start of creating a GenAI environment that is safe and also conducive to building student agency and ownership over their learning. Language is not just interpretative but potentially generative where knowledge acquisition is active not just due to a learner-instructor dimension but through the *individual act of reading* – a faculty that is more active than the mere translation of a particular language through similarity.

4.2. Limitations/future directions

As an extension of the data storytelling framework, a powerful data visualisation tool could be used to augment the Waai experience, in particular to reflect the hierarchical structure in relation to how ideas and concepts may be connected. For instance, a treemap diagram would be able to identify broader categories and subcategories within these. For instance, to illustrate how the feedback output might be reorganised, the treemap might reach into the list of concepts and frame them according to their relationships, thus illuminating their significance and potential for further research. Reflecting both vertical and horizontal dimensions, concepts might be seen as belonging to larger concepts while others are most clearly visualised on different ends of a conceptual spectrum (see Fig. 17).

As demonstrated in the treemap, acknowledging the hierarchy of ideas, which includes their binary relationships, could help a long way in making sense of the lists of concepts offered by the chatbot. In 'feedback' that lists concepts, they may not know, at least in the first instance, which concepts may help to extend their argument. In this regard, key to the implementation of technological affordances such as GenAI tools involves promoting formative feedback (Ndolo, 2021) as an instructional strategy in writing classes in higher learning institutes (Stanier, 2015).

Waai was developed as a result of a belief in asynchronous learning, but how it shapes students' perceptions of writing as an interactive process for the development of self-regulation is through a process of thinking that is made visible. Are the details too literal or do they have a figurative application that helps them move to more conceptual application of ideas? Is their use of language able to move the argument forward, or is there excess description that reveals a need for greater focus in relation to the direction of their writing from their fieldwork observations?

Thus, a framework for design thinking needs to balance exploration and failure with judgement. The output should be transparent enough to reveal what words connotate and not just denote, which is to avoid the assumption that I am able to understand the output in the way I should be able to. Stimulus-organism-response (SOR) theory (Mehrabian & Russell, 1974, p. 8; Huang, 2023) provides a theoretical framework for interpreting user behavior in a progressive manner for an accurate analysis of the effect of GenAI output on the human actor, especially in relation to the unconscious. Thus, a meta-reflection framework (Kesari,

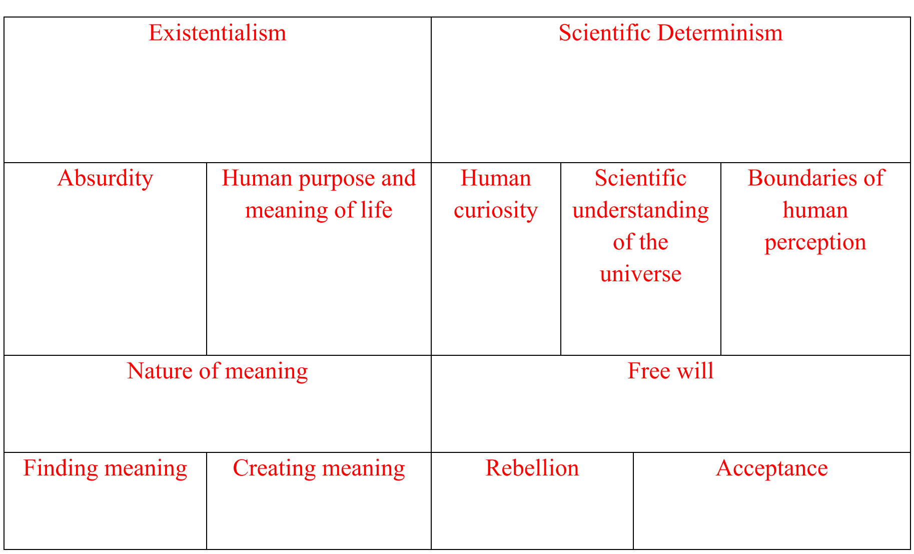

Fig. 17. A treemap visualisation based on P8 feedback output from Nudgy.

2024) is needed to secure observations about student's cognitive engagement with the chatbot (see Fig. 18).

Consequently, the following steps are proposed as a logical sequence for app-making – namely: 1) Identify significance of the output to target audience from the start based on the motivation behind the prompts and features in the app; 2) Clearly guide students in the post-engagement phase so that they know what to do with the chatbot output by distinguishing the "exploratory" and "explanatory" functions of Nudgy (Kesari, 2024); 3) Explore the possibility of conveying the verbal output using a visual diagram to encourage students to dig deeper into concepts generated; 4) Include a meta-reflection phase for students to 'speak back' to the GenAI data output; and 5) Incorporate a guided peer review exercise as part of the brainstorming sequence to reflect on the effectiveness of the prompts used on the Waai interface. Based on the theory of writing as thinking (Roberts, 2022), a design-based digital storytelling as an analytical approach to pedagogical reflection (Long & Hall, 2017) will be used to consider the effectiveness of feedback output generated through LLMs like ChatGPT-4.

Thus, reservations in relation to the use of LLMs concerns what is at stake where writing as a form of expression remains a human endeavour. An authentic voice may be traced to the promotion of thinking that is both divergent and evaluative, especially where critical analysis is not a stable ontology of right or wrong ideas to have, as "teacher presence" used to mean (Stanier, 2015), which Hsieh & Hill, 2022 for instance argues in relation to the centrality of peer review in a writing class. In this way, it becomes necessary to encourage the innovative expansion and contraction of ideas. This is especially crucial in the context of original writing that hopes to eliminate clichés as well as expose arguments that are pseudo-factual. This invisible 'barrier to entry' is assured when AI is used in the classroom, where a physical context exists and may offer opportunities to motivate the curious to apply critical thinking in order to explore creative dimensions of their thinking and writing process. We had asked students if they found the generated phrases to be 'good' – and to their credit, they paused and thought about it. Students trusted that these phrases were 'good' examples. However, when they looked at them, the students' first thoughts were actually something like: If GenAI can think of these phrases, then they are merely conventional – even if they are deemed 'good'. In this way, the capacity for metacognition and reflection is key to the implied pedagogical objective of brainstorming in the context of writing, where it is possible that the intentionality of brainstorming may be a force that could resist the efficiency of prompt engineering as a substitute for intellectual growth.

5. Conclusion

Waai has been built on educational principles that believe in the potential of every student regardless of writing background or ability. It is also developed on the belief that, while tools like ChatGPT may not be an ideal medium for thinking, if used as a stepping stone to motivate students to think more deeply or openly about their subject matter, it may prove to be a hidden prompting hack. By ensuring that the GenAI output is immediate, it could transform writing from a primarily functional purpose to one that is creative and life-changing.

Without taking over the accountability and initiative of students, it is suggested in this paper that the use of GenAI may be justified for educational purposes where it is framed to build metacognitive capacity rather than solely to improve efficiency, which task-based uses of ChatGPT have demonstrated (Hutson, 2021). Moreover, AI-enabled learning avoids the problem of "hallucinations" where, in the case of Waai, ontological validation goes beyond fact-checking. Instead of information being sorted by an external implement, the viability of its instrument involves fine-tuning its built-in meta-prompt design through an ongoing collaboration between instructors and students in the app team.

Like the implementation of critical thinking, human-AI intelligence, or rather augmentation intelligence, deserves a closer look in relation to how collaboration (Chi & Wylie, 2014) and participation in a community of ideas which universities are built on, may be encouraged and enabled by proactive instructional strategy (Aksan et al., 2009). In relation to the pedagogy behind Waai, it is about how original thinking (probably a different kind of knowledge creation than that has been studied in existing pedagogical frameworks thus creating an important point of tension especially where the creative imagination is concerned) could happen for us – both in the classroom and online especially if the use of GenAI has become an inevitable part of our learning experiences. In this way, Waai is a product of the tension between pedagogical and technological design (Kesebir et al., 2010). The underlying issue with motivation was finally that it is no longer sufficient to have produced an asynchronous 24/7 app, even if it is meant to offer personalized insights on student writing. Rather, the focus should be on our students and supporting them as producers rather than just users of content (Yarbrough, 2008).

CRediT authorship contribution statement

Joanne Chia: Writing – review & editing, Writing – original draft, Methodology, Visualization, Validation, Supervision, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Angela Frattarola:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Sophisticated	**Calls for:** Language Prompts that show rather than tell Example: Are phrases suggested effective to increase the immersive nature of the fieldwork experience?	**Calls for:** Examples of good synthesis Example: What makes a good synthesis – is it just finding sources that are different from each other?
Simple	**Calls for:** GenAI chatbot **Example:** Is the prompt output able to target the writing specifically, or generally?	**Calls for:** GenAI chatbot **Example:** Is the prompt output able to identify concepts that directly support the expansion of ideas?
	Exploratory	**Explanatory**

Fig. 18. Data insight approach (Kesari, 2024) for meta-reflection on use of prompts.

Ethics approval obtained

The study was approved by the NTU IRB ethics committee with ID: IRB-2023-384.

Informed consent obtained for research

Informed consent was obtained from all participants, and their privacy rights were strictly observed.

Statement on open data

The data can be obtained by sending request e-mails to the corresponding author.

Original submission statement

The manuscript is unpublished and not under consideration by another journal.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. ASupplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2025.100468.

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